Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Technology development for LMM-based semantic understanding of engineering drawings and generation of structured parametric CAD/process data for downstream manufacturing systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and Materials		Information and communications	
	Medium Category	Small Category	Medium Category	Small Category
	Robot/Automation machinery	IT and software related to robot	Software	SW solution
Concept of Proposed Technology	This technology interprets engineering drawings as *design intent* rather than plain text. Using a Large Multimodal Model (LMM), it understands not only textual content but also the structural and semantic relationships among views, dimension chains, symbols (e.g., GD&T, welding/finishing marks), callouts, and notes. Based on this understanding, the system converts drawing elements into a standardized structured representation (e.g., JSON or a knowledge graph), including entities (parts/features), attributes (dimensions, tolerances, materials), and relations (which note applies to which feature, which dimension belongs to which view/feature, and how constraints propagate across the drawing). These structured outputs are designed to be directly consumed by downstream manufacturing systems and workflows. Typical downstream uses include parametric CAD generation within a predefined template scope, automated extraction of part attributes for BOM/PLM, constraint definition for process planning, and the generation of manufacturing-ready work instructions. To ensure reliability, the system incorporates rule-aware validation (e.g., unit normalization, range checks,			

	cross-view consistency, and constraint compatibility) and produces uncertainty-aware outputs that flag ambiguous cases for review rather than forcing incorrect values. The prototype will initially target standardized product families to secure stable performance and measurable productivity gains. It will then be extended through domain adaptation and human-in-the-loop correction using real industrial drawings and field-defined acceptance criteria, enabling practical adoption in secure on-premise environments where data confidentiality and IP protection are critical.
Rationale for Joint R&D Project	A joint R&D approach is essential for engineering drawing semantic understanding because the problem is simultaneously model-intensive and domain-intensive. On the one hand, robust interpretation of drawings requires advanced multimodal AI capabilities—model architecture design, finetuning strategies, relation reasoning over complex visual–textual structures, and rigorous evaluation protocols. These elements are typically driven by a research partner with expertise in state-of-the-art LMM development and benchmarking. On the other hand, engineering drawings are not a uniform document type: real-world drawings vary significantly by company, product family, and drafting conventions. Critical information is often expressed through proprietary symbol usage, implicit manufacturing constraints, and non-standardized annotation practices. Such characteristics cannot be learned reliably from public datasets alone. Therefore, access to representative industrial drawings, domain rules, and field-defined acceptance criteria—assets typically held by an SME partner—is a decisive factor for practical success. By combining the research partner's technical capability with the SME partner's proprietary data and real manufacturing context, the project can achieve strong synergy. First, domain-specific data and acceptance criteria enable faster convergence to practical performance and reduce ambiguity in schema and requirement definition. Second, iterative validation using real drawings supports early detection of failure modes (e.g., symbol ambiguity, cross-view inconsistencies, dimension-chain mislinking) and enables targeted model and rule-set improvements. Finally, the joint effort accelerates prototype readiness for downstream manufacturing integration, including secure on-premise deployment scenarios where confidentiality and IP protection are critical.
Key Words	1. LMM (Large Multimodal Model) 2. Drawing semantics & Relations Reasoning 3. Design-to-Manufacturing Data Pipeline 4. Work Instruction Automation 5. Domain Adaptation 6. Human-in-the-loop

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	(1) Overall Objectives · Develop an LMM-based semantic understanding engine that interprets engineering drawings as design intent and extracts key entities, symbols/annotations, and relations. · Define and standardize a structured output representation (e.g., JSON/knowledge graph) that captures parts/features, attributes (dimensions/tolerances/materials), and view–callout–feature–constraint relationships. · Implement reliability mechanisms including rule-aware validation (unit normalization, range checks, cross-view consistency, constraint compatibility) and uncertainty-aware outputs for safe industrial use. · Generate structured parametric CAD/process data within a predefined template scope and enable downstream manufacturing integration (e.g., BOM/PLM attributes, process constraints, work-instruction generation). · Deploy an on-premise pilot prototype and validate end-to-end operational effectiveness in real workflows, supported by measurable impact and evidence artifacts (logs/reports). (2) Annual Objectives · **Year 1 (Preliminary research; target TRL 4):** Establish the data/schema/evaluation foundation and deliver a lab-validated baseline pipeline that produces structured semantic outputs from target drawings. · **Year 2 (Integrated prototype; target TRL 5):** Develop an integrated prototype that generates structured parametric CAD/process data within a predefined template scope and demonstrates downstream integration in a controlled test environment. · **Year 3 (Pilot deployment; target TRL 6):** Deploy and operate an on-premise pilot prototype in real workflows and validate operational effectiveness with measurable impact and evidence artifacts.			
R&D Target Specifications	(1) Technical Performance Targets · **Key field extraction F1-score ≥ 0.90 (structured fields):** for critical structured work-instruction fields (material/spec, dimensions, quantities) measured against expert-verified ground truth. · **Process-notes extraction F1-score ≥ 0.80:** for process notes/annotations, evaluated against expert-verified labels after normalization. · **Work-instruction completeness & validity ≥ 90%:** Percentage of generated work instructions that pass required-field completeness and critical rule-based validity checks (unit normalization, range checks, cross-field consistency, mandatory constraints) without major manual correction. (2) Pilot Deployment & Validation Targets · **Processing speed ≤ 60 seconds/page:** End-to-end processing time per drawing page on a single on-premise GPU server (baseline configuration), including work-instruction generation. · **Manual preparation time reduction ≥ 50%:** Reduction in manual work-instruction preparation time during on-premise pilot operation for the target workflow, supported by pilot logs and evaluation reports.			

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R&D Plan	**(1) Research Approach & Methodology (applies across years)** · **Multimodal understanding pipeline:** layout parsing + LMM inference to interpret drawings as design intent (views, dimension chains, symbols, notes). · **Schema-driven structuring:** convert outputs into a standardized work-instruction template and structured representation (JSON/knowledge graph). · **Rule-aware validation & exception handling:** unit normalization, range checks, cross-field consistency, and mandatory constraints, with uncertainty flagging. · **Human-in-the-loop (HITL):** expert review for ambiguous cases, correction logging, and continuous refinement. · **Evidence-based evaluation:** benchmark testing against expert-verified ground truth, plus pilot logs and before/after productivity analysis for field validation. **(2) Year 1 (Preliminary Research) — Target TRL 3–4** · **Annual objective:** Establish the technical/data foundation and deliver a lab-validated baseline that generates standardized work-instruction outputs from target drawings. · **Methodologies:** - **Plan 1:** Define target drawing families, required work-instruction fields, acceptance criteria, and the evaluation protocol (benchmark split and scoring). - **Plan 2:** Build secure ingestion (PDF object + rendering) and establish annotation guidelines and QA procedures. - **Plan 3:** Implement a baseline system for (i) structured-field extraction (material/spec, dimensions, quantities) and (ii) process-notes extraction, producing standardized work-instruction outputs. - **Plan 4:** Implement rule-based validation and an initial exception/error taxonomy to support iterative improvement. · **Milestones:** - Finalized Work-instruction template & schema, labeled/QA-ready dataset, baseline API/demo released - **Accredited test/evaluation institute:** preliminary test plan executed and Year-1 preliminary test evaluation report issued for the baseline system (lab-scale). **(3) Year 2 (Integrated Prototype) — Target TRL 5** · **Annual objective:** Build an integrated prototype for work-instruction and template-scoped CAD/process data generation, and validate it in a controlled environment. · **Methodologies:** - **Plan 1:** Improve model robustness and coverage (symbols/notes, cross-view consistency) through failure-case-driven iteration and domain adaptation. - **Plan 2:** Expand rule/exception handling and implement the HITL workflow (review queue, correction logging, model/rule update loop). - **Plan 3:** Generate structured parametric CAD/process data within a predefined template scope and validate output usability in a test environment. - **Plan 4:** Package the prototype (API, monitoring, logging) and demonstrate controlled-environment downstream integration (BOM/PLM attributes, process constraints, work-instruction automation).

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	· **Milestones:** - Integrated prototype package delivered, controlled-environment integration demo completed, mid-term performance report - **Accredited test/evaluation institute:** Year-2 prototype test evaluation report issued for the integrated prototype in a controlled environment. **(4) Year 3 (Pilot Deployment & Field Validation) — Target TRL 6** · **Annual objective:** Deploy and operate an on-premise pilot prototype in real workflows and validate operational effectiveness with measurable impact and evidence artifacts. · **Methodologies:** - **Plan 1:** Harden and deploy the on-prem pilot (access control, audit logging, monitoring, model/rule versioning) on baseline hardware. - **Plan 2:** Operate the pilot in real workflows; track quality KPIs (structured-field F1, notes F1, completeness/validity) and operational KPIs (latency, throughput, error types). - **Plan 3:** Stabilize performance by reducing manual corrections through failure-case-driven model/rule updates; finalize operational templates and rule sets. - **Plan 4:** Produce TRL6 validation evidence (pilot logs, evaluation reports, before/after impact analysis) and finalize prototype demonstration. · **Milestones:** - On-prem pilot operation completed with evidence (≥50% manual preparation time reduction), pilot logs and evaluation reports, TRL6 validation package - **Accredited test/evaluation institute:** Year-3 field/pilot test evaluation report issued, confirming readiness and performance under on-prem pilot conditions.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	· **Core research leadership (LMM drawing intelligence):** Lead the development of an LMM-based drawing intelligence engine that interprets engineering drawings as design intent, by jointly modeling (i) multi-view layout understanding (projection/section/detail views), (ii) entity extraction (dimensions, tolerances, quantities, material/spec fields, notes), and (iii) relation grounding that links symbols/notes/callouts to the correct geometric entities and view context. The research will explicitly address domain-specific constructs such as dimension chains, weld/finish symbols, GD&T-style annotations, and cross-view consistency requirements. · **Model architecture & training strategy:** Design and refine the end-to-end model pipeline (layout parsing + multimodal reasoning + structured generation), including training/finetuning strategy, domain adaptation, and failure-case-driven iteration to improve robustness across drawing formats and quality levels. · **Structured representation definition:** Specify the standardized work-instruction template and output schema (JSON/knowledge graph), defining required fields, canonical units, value normalization rules, and field-to-field dependencies needed for downstream manufacturing use. · **Evaluation methodology & benchmarking:** Establish ground-truth definition and labeling criteria, build a benchmark protocol (splits, test cases, scoring), implement scoring scripts, and maintain an error taxonomy to diagnose failure modes and guide improvements. · **Validation logic design (technical guidance):** Define rule-aware validation and uncertainty handling strategies (unit normalization, range checks, cross-field consistency, mandatory constraints), including confidence thresholds, “needs-review” triggers, and recommended correction workflows for ambiguous cases. · **Research dissemination (publications):** Publish research outcomes derived from the project (methodology, benchmarking results, and validated findings) in peer-reviewed venues, subject to confidentiality and joint IP/publication agreements. · **Knowledge transfer:** Provide structured knowledge transfer to the SME team, including technical workshops, implementation reviews, reference documentation (model pipeline, evaluation scripts, failure-case taxonomy), and hands-on guidance for deployment readiness. · **IP support for the SME (patents and related rights):** Support the SME in securing intellectual property rights, including invention disclosure drafting support, prior-art search inputs (as available), claim-scope discussions from a technical standpoint, and preparation of technical materials required for patent filing, while ensuring that foreground IP ownership follows the project’s agreed allocation · **Resources to be provided:** Provide research personnel (faculty oversight, graduate researchers), experimental infrastructure and compute resources as available, and participate in regular technical workshops/reviews for design decisions and troubleshooting.

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Expected R&R of Korean SME Partner	· **Data collection & provisioning:** Collect and provide domain engineering drawings (PDF/native exports) and related reference artifacts (existing work instructions, material/spec sheets, cut/process lists). Organize data by product family/drawing type, manage versions, and ensure confidentiality handling and secure transfer. · **Ground-truth creation with domain experts:** Coordinate in-house/field experts to produce and verify ground truth for required work-instruction fields (material/spec, dimensions, quantities, process notes), including edge-case labeling and resolution rules for ambiguous notations. · **Active participation in joint R&D:** Participate in joint research cycles by providing domain knowledge, reviewing intermediate outputs, defining exception cases, and supplying failure cases to drive model/rule improvements. Support periodic technical reviews and co-decision on template scope and acceptance criteria. · **Pilot planning & field validation (demonstration):** Lead on-prem pilot execution in real workflows, including site readiness, user onboarding, KPI logging, and feedback collection. Provide evidence artifacts (pilot logs, before/after productivity measurements, corrected outputs) required to validate operational impact and TRL progression. · **IP protection and ownership:** Identify patentable inventions arising from the project (model/pipeline, validation rules, structured schemas, workflow integration), prepare invention disclosures, and coordinate patent filings and other IP protection actions in line with the agreed IP allocation. · **Commercialization planning:** Define a commercialization pathway for the developed technology, including target market segmentation, product packaging (on-prem software/API/service), pricing/licensing model, deployment/maintenance plan, and customer adoption strategy based on pilot results. · **Operational support for iteration:** Provide practical constraints and engineering feedback to refine the system (workflow integration constraints, runtime requirements, preferred output formats), and support rapid iteration during pilot operation.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**(1) Response to Workforce Reduction Issues and Optimization of Manpower Deployment (Including the Effect of Concentrating Skilled Labor)** · The Document AI-based Outfitting Fabrication & 3D Model Generation system automates the interpretation of complex marine outfitting drawings (e.g., Cable Trays, Supports) and the conversion of design data, thereby reducing the required manpower input. · This lowers reliance on manual work by specific individuals despite structural workforce reduction in the marine equipment manufacturing industry, and mitigates productivity decline even when workforce gaps occur. · It also enables core skilled personnel to focus on high-difficulty judgment, interference checks, and quality approval, improving deliverable reliability and quality. **(2) Improvement of Manual-Based Lead Time Delays (Including Response to Changes)** · By connecting document-based information extraction to fabrication data (BOM, dimension) and 3D CAD generation via an AI pipeline, lead time for generating production drawings is shortened. · When revisions or modification requests occur, the system supports regeneration and re-verification focused on the changed parts, reducing rework scope and shortening re-issuance time. · As a result, recurring bottlenecks at specific stages are reduced, improving overall process speed. **(3) Advancement into a Practical System Based on Joint Research Results with Leading Overseas Institutions (Strengthening Reliability)** · Fine-tune LMM outcomes secured through joint research with leading overseas institutions for the marine outfitting and equipment drawing domain. · Integrate them with a post-processing pipeline (normalization, verification, rule/knowledge combination) to advance into a practical system. · Establish a verifiable automation system that reflects input format deviations, exception cases, and on-site quality standards, lowering deployment risk by strengthening technical reliability. · Contribute to long-term establishment of standardized processes and standard data systems for marine equipment manufacturers and shipyards. **(4) Mitigation of Risks Regarding Skilled Labor Shortage, Variance, and Turnover, and Narrowing the Gap Between New and Skilled Workers (Standardization Effect)** · Standardize criteria through automation of drawing interpretation, extraction, and conversion tasks to reduce output quality deviation driven by individual proficiency. · Enable new workers to reach a stable quality level faster via systemized procedures and verification, shortening the learning curve. · Reduce dependence on skilled workers and limit quality fluctuation and process instability during turnover or shift changes. **(5) Enhancement of Throughput and ROI Based on Mass Processing, Scalability, and Lightweighting** · Rapidly pre-process, extract, verify, and generate large volumes of outfitting component drawings to shorten lead time from review to fabrication start, improving throughput. · Apply model/pipeline lightweighting to process massive quantities of small parts on the same

infrastructure while reducing server/GPU/storage resource usage and operating costs.

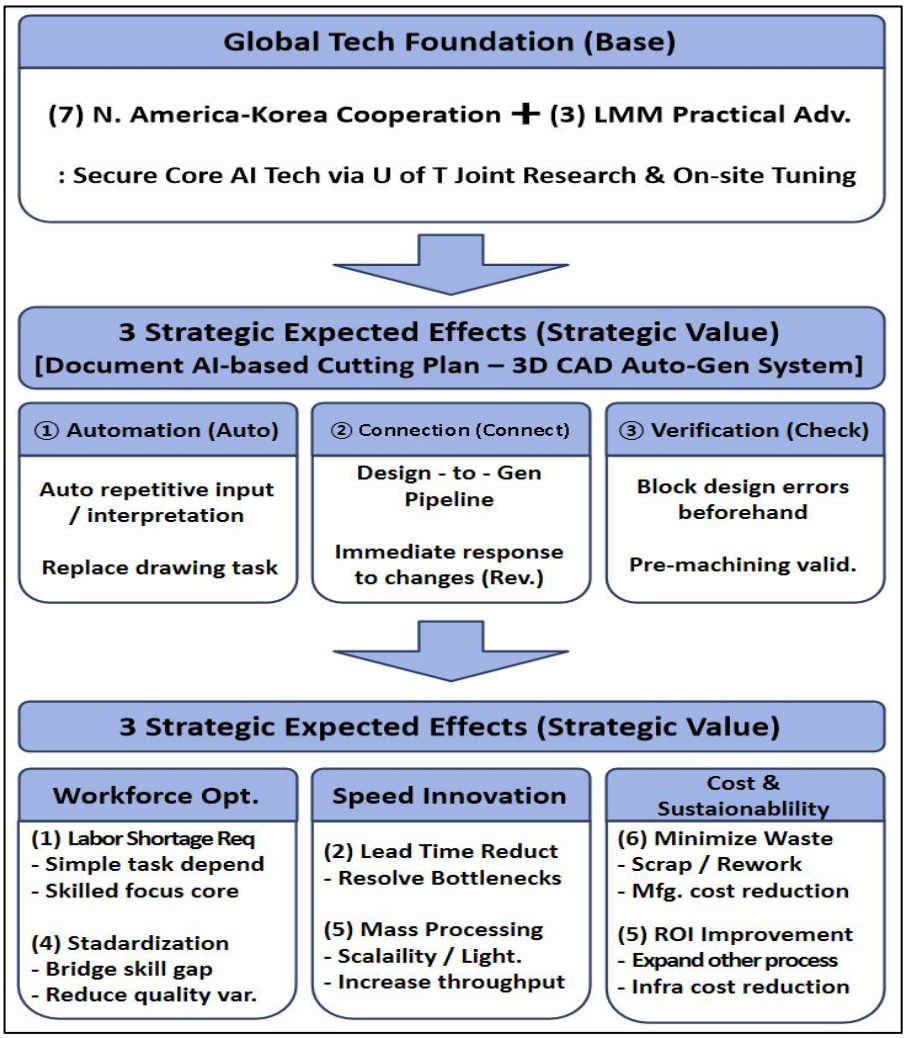
- Reuse development outcomes across projects by extending component-family templates to other outfitting items, broadening applicability and improving long-term ROI.

(6) Preemptive Blocking of Design Errors and Creation of a Resource-Circulating Manufacturing Environment (ESG)

- Human errors in interpreting complex outfitting arrangements often cause costly material waste and installation rework.
- Implement an AI-based automatic verification system to block such errors before fabrication, reducing manufacturing costs and minimizing resource waste.
- Support a sustainable, eco-friendly (ESG) smart process.

(7) Establishment of a Sustainable Advanced Technology Cooperation System Between North America and Korea

- Through joint research with the University of Toronto in Canada, which holds world-class original AI technology, overcome technical limitations faced by domestic SMEs in the specialized marine manufacturing sector.
- Accelerate absorption of advanced technologies and build a durable global R&D cooperation network between Korea and Canada for continuous exchange of cutting-edge methods beyond a one-time project.



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